

right-clicking on it. From Property window, check if all devices have their own footprints.

12. Click on your project name on Navigation window (with extension .dsn). Go to Tool→Design Rule Check. Give options as per needs and click OK. This step will check the design for errors, if any, and those will be shown in the log file. Make necessary corrections.

13. Now open PCB Editor from OrCAD folder on your computer. Got to File→New Drawing. Give a proper name and path (preferably in the folder where you created the project) of the file. Select Board Wizard. The wizard will direct you and give sufficient details in the process. Save the drawings.

14. Come back to the schematic OrCAD window and click on your design file. Then, go to Tools option in the menu and click Create Netlist. The programs communicate using files called netlists.

15. A new window will open. Check Create or Update PCB Editor

and Create PCB Editor Netlist (Fig. 17). Save the input board file as .brd file that you created and saved in the folder. Give a name for the netlist file. Also, select Open Board option on PCB editor. Click OK.

16. PCB editor will automatically open. In Place menu, select Manual option. Now, place all the elements on the side of the rectangular board, and drag and arrange these properly inside the rectangular board place. After arranging the components in a proper manner, without any overlapping, go to Route menu, then PCB router and click on Route Automatic. Routing will be automatically done. You can also do manual routing for better performance (Fig. 18).

17. Go to Manufacture→Artwork. Give the corresponding options. Click on Create Artwork. If you go to the project folder, you will see a file with extension .art. You can print the file also for further manufacturing process.

18. Save your work before closing the window.

Proteus

Proteus ISIS is one of the best simulation software in the world for designing with electronic components including microcontrollers (MCUs). It is a handy tool to test programs and embedded designs. It is a virtual system modelling (VSM) that combines circuit simulation,

animated components and micro-processor models to co-simulate complete MCU based designs.

This program allows you to interact with the design using on-screen indicators, LED and LCD displays, and even switches and buttons. One of the main components of Proteus is Circuit Simulation, using

which you can simulate programming of the MCU. After simulating your circuit using Proteus software, you can directly make PCB design with it. It is an all-in-one package for designers and hobbyists.

To learn Proteus, you can get plenty of tutorials and video; the more you work on it, the better you become. In this section, we describe Proteus 8 Professional version for simulation. Follow the steps given below:

1. Open Proteus by double-clicking the icon on the desktop or navigating to the installed folder (normally the path is C:\Program Files (x86)\Labcenter Electronics\Proteus 8 Professional\BIN\PDS.EXE). Initially, a Home window will be displayed, which shows recent projects, getting-started tutorials, help, latest news and more.

2. To create a new project, go to File→New Project (Fig. 19). It will lead you to New Project Wizard, which helps to create a new project. Give sufficient details like schematic design, PCB layout and firmware. Select default value if you do not know the specifics, or create it later.

3. Finishing the wizard will create a default schematic design just like other EDA tools. Place the components on the schematic window. Placing can be done by selecting Pick Parts From Libraries option in Library menu or by using shortcut key P on the keyboard. This will open Pick Device window.

4. From Pick Device window, either search using the keyword or select the component category-wise (Fig. 20). Proteus supports a large variety of electronic components. Selecting each component shows a schematic and PCB preview with other details. Clicking OK will select the device and an image of that device will be shown when the mouse cursor moves over the schematic window. To place device, move the cursor where you want to and click over the schematic. Use Esc key to

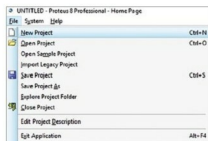


Fig. 19: Creating a project in Proteus

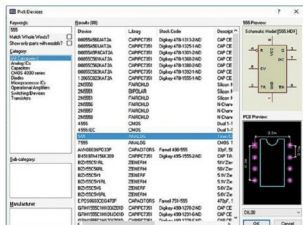


Fig. 20: Pick Devices window